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Key Points:

- A machine-learning model is presented to predict food-security crises in the Horn of Africa
- The model demonstrates high overall performance, and performs similarly to FEWS NET outlooks in the (agro-) pastoral regions
- This study can be utilized to integrate machine learning into existing early warning systems, thereby creating hybrid solutions for the future

Supporting Information:

Supporting Information may be found in the online version of this article.

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Predicting Food-Security Crises in the Horn of Africa Using Machine Learning

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Abstract In this study, we present a machine-learning model capable of predicting food insecurity in the Horn of Africa, which is one of the most vulnerable regions worldwide. The region has frequently been affected by severe droughts and food crises over the last several decades, which will likely increase in future. Therefore, exploring novel methods of increasing early warning capabilities is of vital importance to reducing food-insecurity risk. We present a XGBoost machine-learning model to predict food-security crises up to 12 months in advance. We used >20 data sets and the FEWS IPC current-situation estimates to train the machine-learning model. Food-security dynamics were captured effectively by the model up to 3 months in advance ($R^2 > 0.6$). Specifically, we predicted 20% of crisis onsets in pastoral regions ($n = 96$) and 20%–50% of crisis onsets in agro-pastoral regions ($n = 22$) with a 3-month lead time. We also compared our 8-month model predictions to the 8-month food-security outlooks produced by FEWS NET. Over a relatively short test period (2019–2022), results suggest the performance of our predictions is similar to FEWS NET for agro-pastoral and pastoral regions. However, our model is clearly less skilled in predicting food security for crop-farming regions than FEWS NET. With the well-established FEWS NET outlooks as a basis, this study highlights the potential for integrating machine-learning methods into operational systems like FEWS NET.

Plain Language Summary In the face of increasing droughts and food crises, this study explored the use of machine learning to provide predictions of food crises in the Horn of Africa, up to 12 months in advance. We used an algorithm called “XGBoost,” which we fed with over 20 data sets of potential food security drivers. After training the model, we found that food security dynamics were accurately predicted up to 3 months in advance, especially in pastoral and agro-pastoral regions. The model accurately predicted 20% of crisis onsets in pastoral areas and 20%–50% in agro-pastoral regions with a 3-month lead time. In agro-pastoral and pastoral regions, our machine learning algorithm showed a similar performance to the established early warning system from FEWS NET. The machine-learning model did not show good performance in crop-farming areas. Nonetheless, this study underscores the potential of integrating machine-learning methods into existing operational systems like FEWS NET. By doing so, it paves the way for improved early warning capabilities, crucial in mitigating the looming threat of food insecurity in the Horn of Africa.

1. Introduction

The Horn of Africa is one of the world's most vulnerable regions for food security, with around 57 million people experiencing extreme poverty (UNHCR, 2023). The 2020–2023 drought caused by five consecutive failed rainy seasons was the worst in 40 years (World Meteorological Organization [WMO], 2022). It plunged >20 million people into conditions of high food insecurity and caused acute malnutrition among 7 million children (UNHCR, 2023). Most of the farmers in the region are dependent on rains in the March–April–May (MAM) and the October–November–December (OND) seasons. Meteorological droughts in East Africa have been increasingly observed over the last decades, especially during the MAM season (Funk, Shukla, et al., 2019). Recent literature suggests that these extreme droughts may increase in frequency under anthropogenic warming (Baxter et al., 2023; Funk et al., 2023; Kimutai et al., 2023). Together with expected population growth, this change will further increase the number of food-insecure people over the coming decades (Funk & Shukla, 2020). These trends emphasize the importance of strengthening food-security early warning systems and increasing the understanding of potential drivers of food-security crises in different contexts. Early action based on early warning

systems save lives and financial resources compared to late response, as was shown by the 2011 drought in the Horn of Africa (Dempsey & Hillier, 2012).

Currently, several drought early warning systems with a focus on food security are operational, such as the Hunger Hotspot early warnings from the World Food Program (WFP) and the Food and Agriculture Organization of the United Nations (FAO) (WFP and FAO, 2022), and the FAO Global Information and Early Warning System on Food and Agriculture (GIEWS) (FAO, 2023b). The most widely used early warning system for Africa is the U.S. Agency for International Development's Famine Early Warning Systems Network (FEWS NET). FEWS NET uses a combination of observed and forecasted drought indicators, vulnerability indicators and expert judgment to make local and regional assessments of food security. This process is conducted using key Integrated Food Security Phase Classification (IPC) protocols (IPC, 2023). These assessments not only pertain to the food-security assessment of the current situation but also include projections of food security for the near term (up to 4 months in future) and medium term (up to 8 months in future). These projections are summarized in Food Security Outlooks (FEWS NET, 2023b). Together with national partners, humanitarian agencies such as the Red Cross and Red Crescent Movement and WFP have introduced anticipatory action for food insecurity and drought over the last decade (WFP, 2023a). Food-security outlooks and reliable early warning signals are crucial to triggering these anticipatory actions.

Over last decades, many advancements have been made in improving food security forecasts. However, significant gaps still persist. First, the main food security early warning system, FEWS NET's food-security outlooks, face challenges in predicting food security (Backer & Billing, 2021; Krishnamurthy et al., 2020). Although soil moisture and drought forecasts (Boult et al., 2020; Shukla et al., 2020) as well as crop yield forecasts (Lentz et al., 2022) can often be accurate weeks to months in advance, food-security dynamics are often hard to predict because of region-specific or unexpected drivers (e.g., conflict or desert-locust outbreaks). Second, studies testing the added value of alternative prediction methods, such as machine learning, are still scarce. The recent emergence of machine learning in predictive analytics and early warning systems has been shown to enhance disaster mitigation and preparedness (Ghaffarian et al., 2023). These systems make use of the explosion in available data and new analytical methods (Lentz & Maxwell, 2022), such as the XGBoost machine learning algorithm (Chen & Guestrin, 2016). Promisingly, the existing studies demonstrate that machine learning holds the potential to efficiently monitor (Martini et al., 2022) and predict (Backer & Billing, 2024; Balashankar et al., 2023; Foini et al., 2023; Lentz et al., 2019; Westerveld et al., 2021) food consumption, malnutrition, and food insecurity by utilizing data on their drivers. However, many facets of machine-learning predictions of food crises remain unexplored. To our knowledge, none of the existing studies focus on the entire Horn of Africa region. Westerveld et al. (2021) developed machine-learning predictions for Ethiopia, but predicted food security transitions instead of the FEWS IPC value itself. Furthermore, most of the existing studies lack explainable machine-learning techniques to shed light on the underlying potential drivers of food security on various lead times (e.g., Balashankar et al., 2023; Foini et al., 2023). Moreover, existing machine-learning models are rarely compared to existing operational systems, such as the food security outlooks from FEWS NET. This is vital to an assessment of the added value such machine-learning approaches can bring. Bridging these two gaps is essential to understanding the potential of machine-learning algorithms in different contexts, and to understand how they can be adopted and integrated into the currently used consensus-based approaches.

Consequently, the primary objective of this study is to predict food security dynamics and crisis situations in the Horn of Africa months in advance using historical data on natural and socio-economic processes in a machine-learning model. This study will increase the understanding of lead-time-dependent food-security drivers and reveal the performance of these machine-learning models. We will compare our machine-learning predictions with FEWS NET food-security outlooks to identify where our predictions can provide additional value.

We begin with an outline of the methodological framework (Section 2). In the results (Section 3), we outline the forecast accuracy and the use of explainable machine-learning techniques to elucidate the underlying dynamics of food-security drivers. Subsequently, we consider these results in the discussion (Section 4) and provide our main conclusions and recommendations (Section 5).

2. Methods

Figure 1 shows the setup of our approach to predicting food security in the Horn of Africa, which we refer to here as encompassing the territories of Kenya, Somalia and Ethiopia. We use the XGBoost model (Chen &

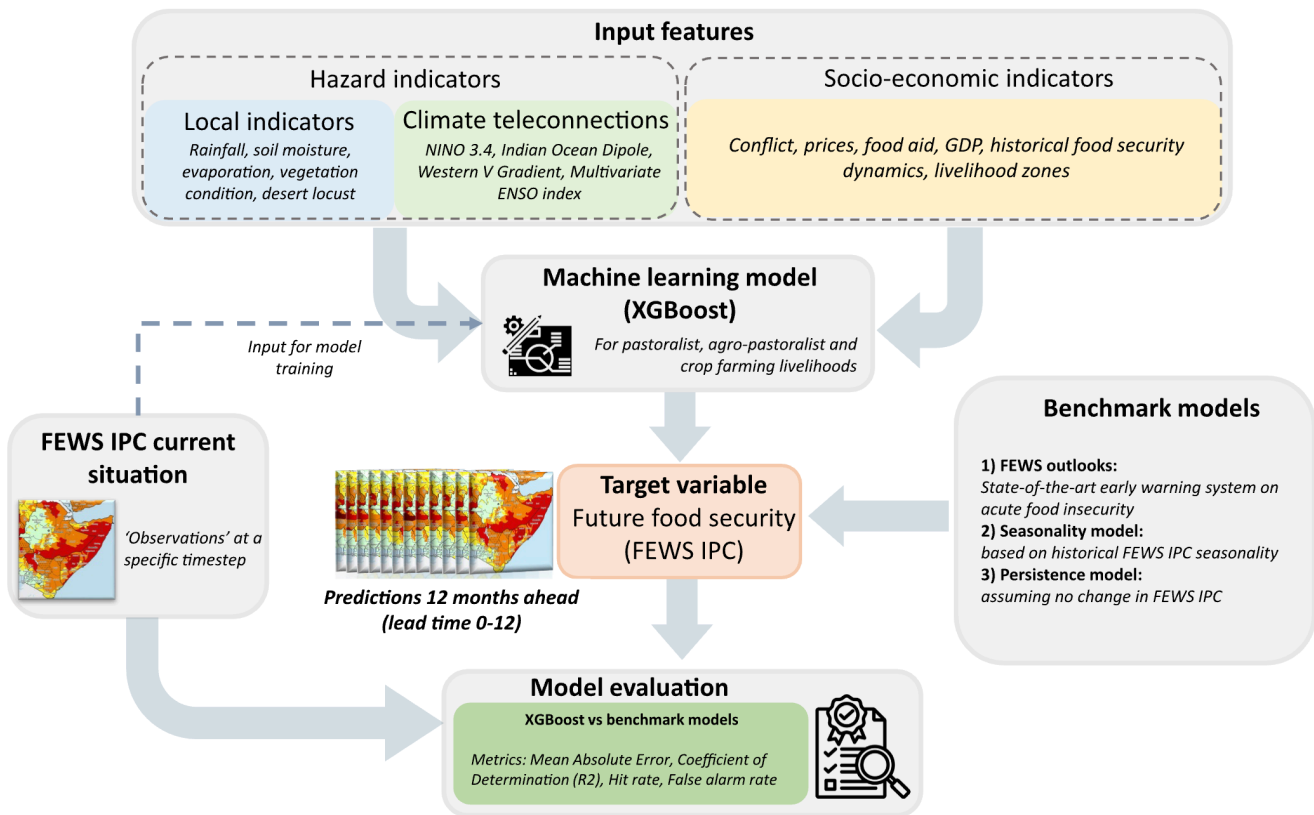


Figure 1. This study design shows the input data employed in the modeling framework, which includes features used (top) and the FEWS IPC current-situation maps (left). These elements feed into the machine-learning model (XGBoost) and are utilized to train the model and make predictions of future food security (center). We compare these predictions to the current situation to evaluate the machine-learning model (bottom). Predictions are also made using three benchmark models (right), one of which is the state-of-the-art outlook from the FEWS NET early warning system.

Guestin, 2016) as our machine-learning model to predict IPC food-security status on monthly and seasonal timescales (Section 2.1.1). XGBoost, or Extreme Gradient Boosting, is an ensemble decision-tree algorithm, like random forest regressions, but able to model more complex interactions due to its ability to boost individual trees. We add multiple explanatory natural hazard and socio-economic variables (Section 2.1.2) as potential drivers for food insecurity, which we refer to as “features.” Subsequently, the XGBoost model (Section 2.2.1) is trained (2009–2019) to predict FEWS IPC food-security states on multiple lead times. These predictions are tested on separate “hold-out” data set (2019–2022) of FEWS IPC values (Section 2.2.2). In this stage, three benchmark models are used to which our predictions will be compared: (a) the FEWS NET outlooks, (b) a persistence model (prediction same as now), and (c) a seasonality model (Section 2.5). These steps are visualized in Figure 1.

2.1. Data Overview

The modeling framework aims to predict FEWS IPC acute food-insecurity values (target variable; Section 2.1.1). To make these predictions, 20 input variables (hereafter referred to as “features”; Section 2.1.2) are included. We used administrative units as the spatial level of the analysis as they have a more consistent and generally smaller size than other spatial units, such as livelihood zones. Moreover, they better facilitate the translation of our research into early warning and action practices and policies. The mean of each feature and the FEWS NET data were calculated for each administrative unit (admin-1 or -2 levels). For Somalia and Ethiopia, the admin-2 level was used, but for Kenya, the admin-1 level was used, as the admin-2 regions (290 in total) resulted in an impractical level of spatial differentiation for the modeling. This resulted in a total of 213 administrative units included in the study.

2.1.1. FEWS IPC Food-Security Outcomes

As described above, the FEWS IPC food-security maps are chosen as the target variable to be predicted with the model. We first provide background information on how the FEWS IPC food-security status is determined, after which we outline how the data was pre-processed to be used in the model.

2.1.1.1. Acute Food-Insecurity Monitoring Using IPC

Global food-security assessments consist of four main pillars: food access, food availability, food utilization, and stability (FAO, 2009). The Integrated Phase Classification (IPC, 2023) was developed to represent and evaluate these pillars. The IPC, however, uses three different scales to measure food security and nutrition: acute food insecurity, chronic food insecurity, and acute malnutrition.

In this study, we focus on acute food insecurity. The IPC estimates the magnitude of acute food insecurity and identifies its key drivers (IPC, 2021, Figure 27). Acute food insecurity is measured using internationally recognized scientific standards and cut-offs on a five-phase scale: Phase 1, minimal/none; Phase 2, stressed; Phase 3, crisis; Phase 4, emergency; and Phase 5, catastrophe/famine (IPC, 2021). This system defines first-level food-security outcomes—food-consumption gaps and negative livelihood change—which, if the situation worsens, result in second-level acute malnutrition and mortality outcomes. The combination of these outcomes determines the IPC classification, as malnutrition and mortality can also be caused by factors other than food-consumption gaps. Negative livelihood change is an important indicator because unsustainable livelihood practices, such as the reduction of health expenditures or risky migration, temporarily decrease food-consumption gaps but strongly increase long-term vulnerability. Therefore, IPC can, in this case, still assign a high acute food insecurity class (IPC, 2021).

2.1.1.2. The “Target Variable”: FEWS IPC Food-Security Maps

FEWS NET uses the IPC scale and key protocols described above to produce the FEWS IPC acute food security estimates (FEWS NET, 2024). The estimates of the current situation provided by FEWS NET—hereafter referred to as FEWS IPC values—serve as the primary target variable for our machine-learning model, as illustrated in Figure 1 (left). The maps are downloaded as shapefiles from the FEWS NET data portal (FEWS NET, 2023a). The FEWS NET current situation estimates are released three to four times per year, depending on the period considered (see Backer & Billing, 2021, Figure A1). We used area-level classifications, which assign the highest food-security class faced by at least 20% of the population. From these maps, we calculated the population-weighted spatial mean per administrative unit. We used the gridded WorldPop (2024) unconstrained data adjusted to match the official country totals from the United Nations. A population-weighted spatial mean is most appropriate for the aggregation to administrative units, as the FEWS IPC levels are population-based.

2.1.2. The “Features”: Potential Drivers of Food Insecurity

We used a total of 20 features in the models (Table 1), which are classified into natural hazard indicators (local hazard indicators and climate teleconnections) and socio-economic indicators. The natural hazard indicators, mostly consisting of rainfall and drought indicators, are chosen due to the strong link between drought and food security (Funk, Shukla, et al., 2019; Svoboda & Fuchs, 2017), including a strong influence of climate teleconnections (Funk et al., 2023). The socio-economic indicators are selected using insights from WFPs Vulnerability Analysis and Mapping (VAM) system (WFP, 2023b) and the existing studies on food security forecasting. The features were all aggregated to the scale of administrative units with methods described below.

2.1.2.1. Natural Hazard Data

Rainfall indicators: We used daily CHIRPS rainfall data (Funk et al., 2015) over the period 1981–2022. We calculated total rainfall, total number of wet days (>1 mm/day), and maximum dry-spell length (>5 consecutive dry days) per month. The maximum dry-spell length can be > 31 consecutive days if the dry spell extends over several months. We averaged these indicators to the scale of administrative units.

Drought indices: We included three different drought indices that are pivotal for objective drought monitoring at different spatial and temporal scales: the standardized precipitation index (SPI) (McKee et al., 1993), the standardized precipitation evapotranspiration index (SPEI) (Vicente-Serrano et al., 2010), and the standardized soil

Table 1

The Features in the Model, Including Local Natural Hazard Indicators, Climate Teleconnections, and Socio-Economic Indicators

Natural hazard indicators					
Local hazard indicators		Climate teleconnections		Socio-economic indicators	
Name	Source	Name	Source	Name	Source
Total rainfall	CHIRPS; Funk et al. (2015)	Indian Ocean Dipole (IOD)	NOAA (2023b)	ACLED: number of conflicts and fatalities	Raleigh et al. (2010)
Number of dry spells	CHIRPS; Funk et al. (2015)	Multi-variate ENSO index (MEI)	NOAA (2023a)	Food and fuel prices	WFP (2023b)
Number of wet days	CHIRPS; Funk et al. (2015)	NINO3.4	NOAA (2023a)	Historical and current food-security situation	FEWS NET (2023a)
Standardized precipitation index (SPI)	CHIRPS; Funk et al. (2015)	Western V gradient (WVG)	Funk et al. (2023)	Humanitarian food assistance	FEWS NET (2023a)
Standardized soil moisture index (SSMI)	Martens et al. (2017)			Headline and food consumer price index	Somalia (NBS, 2023). Kenya and Ethiopia (Ha et al., 2021)
Standardized precipitation and evaporation index (SPEI)	Funk et al. (2015); Martens et al. (2017)			Gross domestic product (GDP) per capita	IMF (2023)
Normalized difference vegetation index (NDVI)	NOAA (2021)				
NDVI croplands	NOAA (2021); Pérez-Hoyos (2018)				
NDVI rangelands	NOAA (2021); Pérez-Hoyos (2018)				
Desert-locust swarms	FAO (2022)				

moisture index (SSMI) (Blauhut et al., 2016; Hao et al., 2014). The SPI measures meteorological drought conditions and is based on CHIRPS. To include wider atmospheric conditions, we calculated the SPEI, which is the standardized difference between precipitation and potential evapotranspiration (PET). The PET was retrieved from the global land evaporation Amsterdam model (GLEAM) (version 3.5a; Martens et al., 2017) and reflects atmospheric conditions such as wind speed, temperature, and humidity. The GLEAM model uses satellite and reanalysis data to estimate land-surface evaporation and soil moisture on a 0.25-degree grid. Additionally, the SSMI drought indicator is derived from GLEAM using the root-zone soil-moisture data set.

The above drought indices are calculated using the methodology described in Odongo et al. (2023). Specifically, the monthly indices are derived by accumulating the input variables (rainfall for SPI, rainfall and PET for SPEI, and root-zone soil moisture for SSMI) over 1, 3, 6, 12, and 24 months. Subsequently, a distribution was fitted through the accumulated variables, and the data was standardized by comparing these variables with the amount of the variable that would have been expected based on the long-term climatology (1981–2022). For the standardization, we used a statistical distribution that best fitted the data between -3 and $+3$. Multiple distributions were tested for each of the indices per period and administrative unit. The distribution with the best fit based on the Kolmogorov best-fit test was selected (see Odongo et al., 2023, for details). The calculation of the SPI was corrected for zero values in the distribution, as recommended by Stagge et al. (2015). Averaging was used to aggregate the indices to the scale of administrative units.

Agricultural indicators: We included NDVI as the agricultural drought indicator, derived from the NOAA Satellite Center for Satellite Applications and Research (NOAA, 2021). This data set contains data from the Advanced Very-High-Resolution Radiometer (AVHRR) sensor. The archive contains validated 7-day composites of smoothed NDVI data at 4 km² resolution. We extracted rangeland NDVI and cropland NDVI values using cropland and rangeland masks from the Anomaly Hotspots of Agricultural Production (ASAP) system (Pérez-Hoyos, 2018). Subsequently, the NDVI values were expressed as anomalies per month using 2000–2021 as the reference period, and averaged over the administrative units.

Desert locusts: We included over 10,000 data points on swarms of desert locusts obtained from the FAO Locust Hub (FAO, 2022). The total area affected each month was calculated as a percentage of the overall area within the defined administrative division.

Climate teleconnections: The climate in the Horn of Africa is strongly influenced by sea surface temperature (SSTs) in the Indian and Pacific Oceans (Funk et al., 2023). Therefore, we included multiple variables representing these SSTs: the Indian Ocean Dipole (IOD) (NOAA, 2023b), the multivariate ENSO index (MEI), and NINO 3.4 ((NOAA, 2023a). Recent research has also discovered a new gradient in the Pacific Ocean called the Western V gradient (WVG), which is linked to the severe drought conditions in East Africa observed over the past several years (Funk et al., 2023). Consequently, we included the WVG as observed during the MAM season. We assigned all SST indices to all administrative units. The areas used for the calculation of the SST indices are visualized in Figure S1 in Supporting Information S1.

2.1.2.2. Socio-Economic Data

Food and fuel prices: We used food and fuel prices from the WFP's price database, the VAM Food Security Portal (WFP, 2023b). We selected maize as the main food crop for each country based on information from the Global Information and Early Warning System from FAO (FAO, 2023a). Other crop prices were not included due to limited data availability in WFP's price database. We also included fuel prices (diesel), as this was an important driver of past food crises (WFP and FAO, 2022). We used the Alert for Price Spikes (ALPS) indicator (WFP, 2014) as a means to obtain standardized prices that are corrected for the long-term (seasonal) trend. Using this indicator, WFP aims to detect price spikes and abnormal price deviations beyond long-term trends. Details of the ALPS-indicator calculation can be found in WFP's ALPS manual (WFP, 2014).

The data is provided for 14 markets in Kenya, 98 markets in Ethiopia, and 29 markets in Somalia. We geolocated these markets and subsequently selected the closest market for each administrative unit. For each time step, data gaps in the closest market for a specific administrative unit are filled by the closest market in the country for which data is available.

Macroeconomic indicators: We include inflation using data from the National Bureau of Statistics (NBS) for Somalia (NBS, 2023) and the World Bank Global Inflation Data set (Ha et al., 2021) for Kenya and Ethiopia. This includes the consumer price index (CPI) as a measure for overall inflation ("headline CPI") and inflation on food products specifically ("food CPI"). Furthermore, we used the gross domestic product (GDP) per capita as an indicator for national economic growth (IMF, 2023). Incorporating these national-level indicators can enhance the model's performance, especially given that the machine-learning models are trained on pooled data encompassing administrative units from all three countries, yet within the same livelihood zone as detailed in Section 2.2.1. The indicators for each country were assigned to the country's administrative units.

Historical food-security situation: Upcoming food-security dynamics are dependent on current and past food security situations. For example, low food-insecurity stages more easily transition to high food insecurity than vice versa (Wang et al., 2020). XGBoost can learn such relationships, so we include the FEWS IPC values of the previous timesteps as features in the model. These values, including the humanitarian food assistance, were aggregated to the administrative units using the same population-weighted averaging as deployed on the FEWS IPC current situation (the target variable, see Section 2.1.1).

Humanitarian food assistance: Data on the impact of humanitarian food assistance (recorded since 2012) have been extracted from the FEWS NET data portal (FEWS NET, 2023a). Such aid includes direct food assistance (i.e., in-kind food transfers) but may also include indirect food assistance (i.e., cash or livestock assistance). The data marks areas that would likely have been one phase more food insecure without significant humanitarian food assistance (FEWS NET, 2023a), which are indicated with an exclamation mark (!) in the food-security map published by FEWS NET.

Conflicts: Conflict data was extracted from the Armed Conflict Location & Event Data Project (ACLED; Raleigh et al., 2010) data set. Since 1997, the ACLED data set has collected events of political violence and protest across 50 states worldwide, such as those resulting from rebels, governments, or militias. Each entry represents a single event of a specific type at a particular location on a given day. We calculated both the total number of conflicts and the total number of fatalities per administrative unit per month.

2.2. The Machine-Learning Model Architecture

2.2.1. The XGBoost Model

Decision-tree models have shown great potential for impact-based forecasting of climatic shocks (Everingham et al., 2016; Guimarães Nobre et al., 2019; Schoppa et al., 2020; Westerveld et al., 2021). In our study, we selected XGBoost (eXtreme Gradient Boosting) as the regression model of choice (Chen & Guestrin, 2016; Friedman, 2001). XGBoost is an advanced model that improves upon traditional decision tree approaches by not relying on a single tree. Instead, it integrates an ensemble, or group, of different decision trees. It uses a scalable tree-boosting system to optimize predictions. Simple decision trees, which are called shallow (weak) learners, are iteratively added to minimize the errors of previous predictions. Simultaneously, these trees are subjected to regularization, which is a technique to prevent them from overfitting. This makes the XGBoost algorithm distinct from other ensemble decision-tree based models, such as random forests.

We selected this model due to its demonstrated speed and more effective performance compared to other models in many different fields (Chen & Guestrin, 2016), including drought and food-security prediction (Foinin et al., 2023; Martini et al., 2022; Westerveld et al., 2021; Zhang et al., 2023). The tree-boosting systems allow XGBoost to better model complex and non-linear relationships, which are often present in food-security systems.

A different XGBoost model was created for each lead time (0, 1, 2, 3, 4, 8, and 12 months), which resulted in seven separate models. The data from the 213 individual administrative units was pooled in three livelihood zones (Figure 3): (a) pastoral, (b) agropastoral, and (c) crop farming. In case a administrative unit contained multiple livelihood zones, we selected the livelihood zone that covered the majority of the area. A different model was made for each livelihood zone, which, combined with the different lead times, resulted in 21 unique machine learning models. The pastoral and crop-farming regions are the largest, with 82 and 106 administrative units respectively, whereas the agro-pastoral regions consist of 25 different units.

2.2.2. Machine Learning Setup: Train–Test–Validation

Figure 2 illustrates the setup of the machine learning models. For each model, the time series data are divided into a training set and a test set, adhering to an 80:20 ratio based on time. This results in a training data set from 2009 to 2019 (Figure 2, blue line) and a test data set from 2019 to 2022 (Figure 2, red line). The time series presented are for one of the 213 administrative units (Mandera in Kenya). The test data set is out-of-sample, meaning that we leave it untouched and only use it for model testing. An out-of-sample approach is a beneficial practice in time series forecasting to ensure temporal independence of the data set (Cerqueira et al., 2020). We did not shuffle the observations prior to the train-test splitting, because maintaining the original temporal sequence of observations is crucial for time series data (see for example, Cerqueira et al., 2020; Snijders, 1988).

2.2.3. Hyperparameter Optimization

We executed a walk-forward cross validation to tune the hyperparameters of the model, using 5 different validation sets (Figure 2b). We found the following optimal hyperparameters: maximum tree depth: 4, number of trees/estimators: 400 and learning rate: 0.01 (see Table S1 in Supporting Information S1). Extended information on the hyperparameters and the tuning process can be found in Supporting Information S1 (Text S1). The evaluation scores on the five validation sets (i.e., the five different folds) are shown in Table S2 in Supporting Information S1.

2.3. Feature Engineering

We further process the features from the data series listed in Section 2.1 using feature engineering (Zheng & Casari, 2018). This brings time and memory effects to the XGBoost model. For each input feature, we compute the rolling average over both the last 4 and 12 months for each individual month. These longer accumulations can be important to the model, as food security crises often develop over longer timescales such as during the 2020–2023 drought (WMO, 2022). We identified the timing of the main rainy seasons in the Horn of Africa, MAM and OND, which are marked as “1” in the model data set. Memory effects in the target variable FEWS IPC were accounted for by including values from 1, 4, and 8 months prior, along with the mean FEWS IPC value from the past 12 months. Additionally, we incorporated country names into the model, enabling it to factor in country-

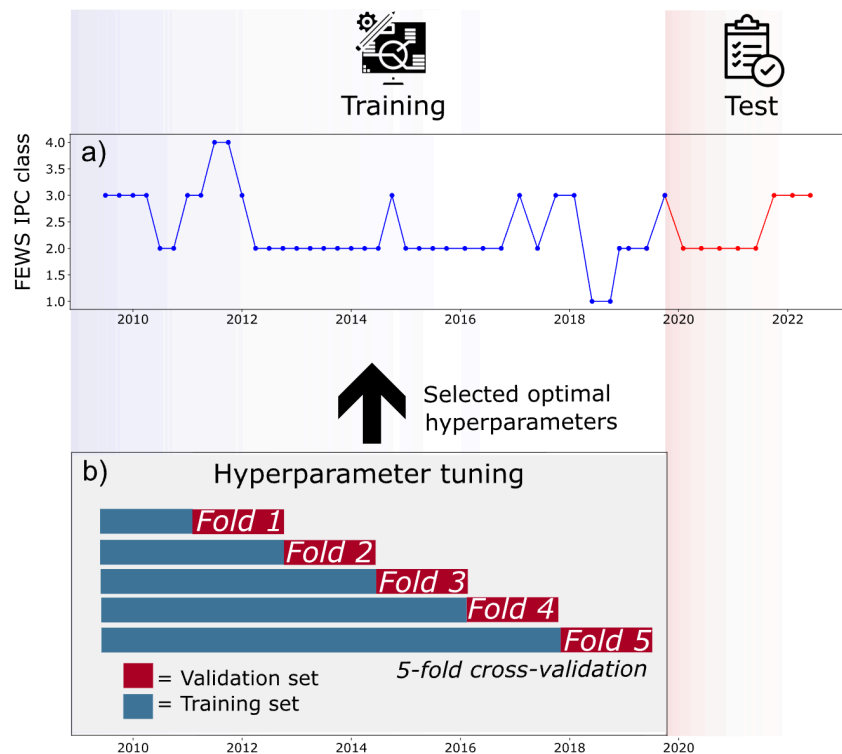


Figure 2. (a) The setup of the machine-learning model splits the features and target variables into training and test parts. As an illustration, the time series shown represents the FEWS IPC values for Mandera in Kenya. (b) The training part is also used for hyperparameter tuning of the model using a 5-fold walk-forward cross validation.

specific elements (such as drought-intervention policies) not included in the original data. This resulted in a total of 80 unique features.

2.4. Consideration of Lead Time

Predictions were made for various lead times (0, 1, 2, 3, 4, 8, and 12 months) using seven distinct XGBoost models. Each model was trained with specific lags corresponding to the respective lead time. The lags were implemented in the model by using features from the past to predict the target variable FEWS IPC in a specific time step. To achieve these lags, the features were shifted in time by an amount of months equal to the lead time before model training started. Thus, these time shifts take place for all lead times, except for a lead time of 0 months. These lags allow the model to learn the relationships between features and FEWS IPC classes separated by the prediction lead time.

Predictions are only generated for months when the FEWS IPC observation is released, which occurs three to four times annually. When predicting the food-security outcome with a 1-month lead time, the features utilize data recorded in the “current” month and the preceding months to make a forecast for the next month. This “historical” data is integrated through feature engineering, as outlined in Section 2.3.

2.5. Benchmark Models and Performance Metrics

The predictions in the test set (2019–2022) were evaluated using the mean absolute error (MAE), the coefficient of determination (R^2), the hit rate, and the false-alarm rate. Three benchmark models were used and served as performance reference: (a) the state-of-the-art *FEWS NET* food-security outlooks, (b) a *seasonality model* based on historical FEWS IPC observations, and (c) a *persistence model* assuming no change in the FEWS IPC class. The seasonality model makes predictions using the monthly average FEWS IPC value for each administrative unit, as calculated over the training period (2009–2019). Every prediction with the seasonality model is similar for different lead times because they all use the seasonality from the training set. The persistence model relies on the

last-observed FEWS IPC value for making predictions, which is released three times a year over this period. However, for lead times beyond 3 months, the model cannot use the last value. Instead, it must utilize the FEWS IPC observation before it. For example, we assume that the FEWS NET current-situation release occurs by the end of the month, and therefore the persistence predictions for October on lead 0 cannot make use of the FEWS NET release in that month. The FEWS NET food-security outlooks require a more detailed explanation, which we provide below.

2.5.1. The FEWS NET Food-Security Outlooks

The FEWS NET food-security outlooks are state-of-the-art projections from FEWS NET. They are the result of a rigorous scenario-development process, which leads to a “most likely” future food-security scenario (FEWS NET, 2018). The outlooks utilize different information sources, such as rainfall and temperature observations, but also climate modes, including ENSO. At lead times of 3–6 months, FEWS NET uses long-range seasonal forecasts, such as root-zone soil moisture (Shukla et al., 2020). Local vulnerability is incorporated through knowledge and experience of livelihoods, market dynamics, and nutrition (WMO, 2017).

Two types of food-security outlooks are created through FEWS NET for the coming 8 months: near term (the first 4 months) and medium term (the second 4 months) (FEWS NET, 2018). These outlooks, released every month, target the month(s) just prior to the FEWS NET current-situation observations. To validate the outlooks, we compared them to the next available current-situation observation in February, June, or October.

2.6. Interpretation of Model Results

Machine learning is often criticized as a black box (McGovern et al., 2019), which emphasizes the need to increase model transparency. Therefore, we use the SHAP (Shapley Additive Explanations) (Lundberg & Lee, 2017) framework to interpret model predictions and understand how the model uses the features. Shapley values originate from game theory (Shapley, 1953) and are solutions to the problem of dividing a game's single payout among all players according to their respective contributions. In this case, the payout is the prediction of the statistical model, and features are the contributors. This framework is unique in the sense that it shows the impact of every individual feature on each prediction, which is also called “local feature importance.” The SHAP values for every input feature reveal how that feature changed the model prediction at that specific time step. The SHAP values are in the same unit as the target variable, so that a SHAP value of 0.4 means that this feature increased FEWS IPC food insecurity with 0.4. Thus, SHAP can reveal the influence of any of the features on any prediction. This differentiates SHAP from the many other explanation methods based on global interpretation that only show the contribution of the features to the model as a whole. Nonetheless, combining all local SHAP values provides a realistic view of global feature importance (Lundberg et al., 2020).

3. Results

3.1. General Model Evaluation

The model is evaluated using the test data set covering the period from 2019 to 2022. This unique period includes the 2020–2023 drought and food crises (WMO, 2022). This makes the period highly suitable for testing the model's ability to predict food crises. An illustration of the predictions with the observations demonstrates the ability of the XGBoost model to predict food-security dynamics over different administrative units in the region (Figure 3). This includes (a) the correct timing of the onset of crisis in, for example, Burco, Waajid, and Garissa and (b) the dynamics of phases in low food security (e.g., in Tana River). Both the timing and dynamics are predicted effectively, with R^2 values up to 0.94 in Afmadow (Somalia).

We continue with an in-depth model evaluation based on the test data set. Figure 4 shows an assessment of the quality of the 3-month forecasts with the XGBoost model across the Horn of Africa region. They have an average MAE (mean absolute error) of 0.35. This is acceptable compared to the standard deviation of the observed FEWS IPC values (0.8). The Tigray region in northern Ethiopia is a clear outlier with poor skill. The region experienced a sudden increase in food insecurity during our test period (2019–2022) resulting from the outbreak of armed conflict in November 2020. Conflicts are included in the model through the ACLED data set. However, the relationship between conflict and food insecurity is complex (see Section 4.3.3) and led to a large underestimation of food insecurity in Tigray (see Figures 3 and 4, Western Tigray).

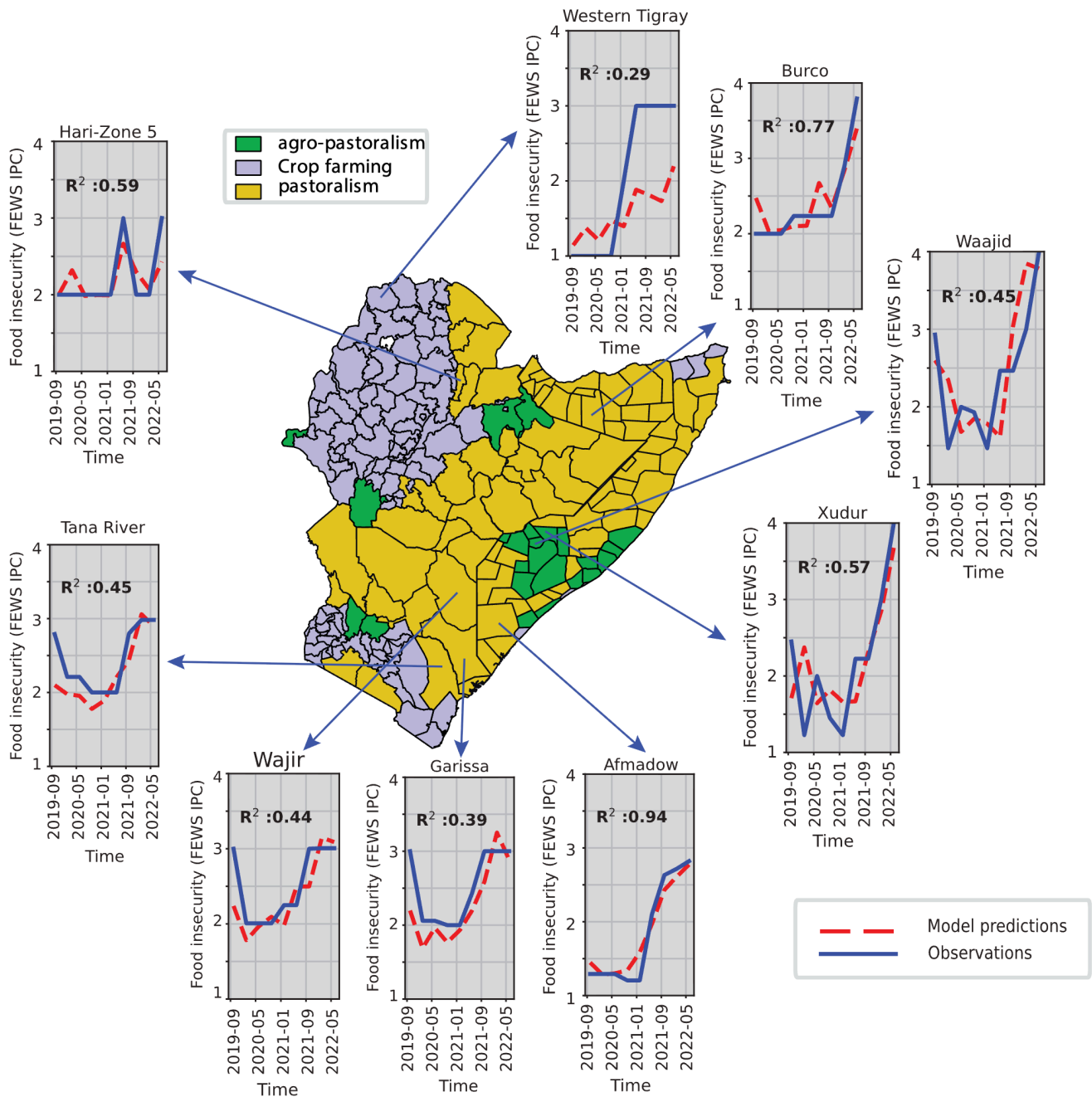


Figure 3. Time series show XGBoost model predictions of FEWS IPC food insecurity with a lead time of 3 months (red lines), compared to observations (blue lines). Examples of the different administrative units are selected over various regions and livelihood zones. The map displays the three individual livelihood zones for which the machine-learning models are trained.

The XGBoost model dynamics were further evaluated using the coefficient of determination (R^2). Over the whole region, high R^2 values were found on short lead times ($R^2 = 0.72$ on 1 month lead; Figure 5d). However, the model systematically underestimated high FEWS IPC values of 4. We also assessed the model performance in different livelihood zones (Section 2.2.1) as compared to the benchmark models (Figure 5, top). The XGBoost model demonstrates effective performance, with $R^2 > 0.5$ recorded for predictions less than 4 months in advance (Figure 5, green line). For longer leads, the performance decreases. Our model outperforms the persistence and seasonality benchmark models for all three livelihood zones, especially in agro-pastoral and pastoral regions. However, the performance of these baseline models is significantly better in crop-farming regions, which may be

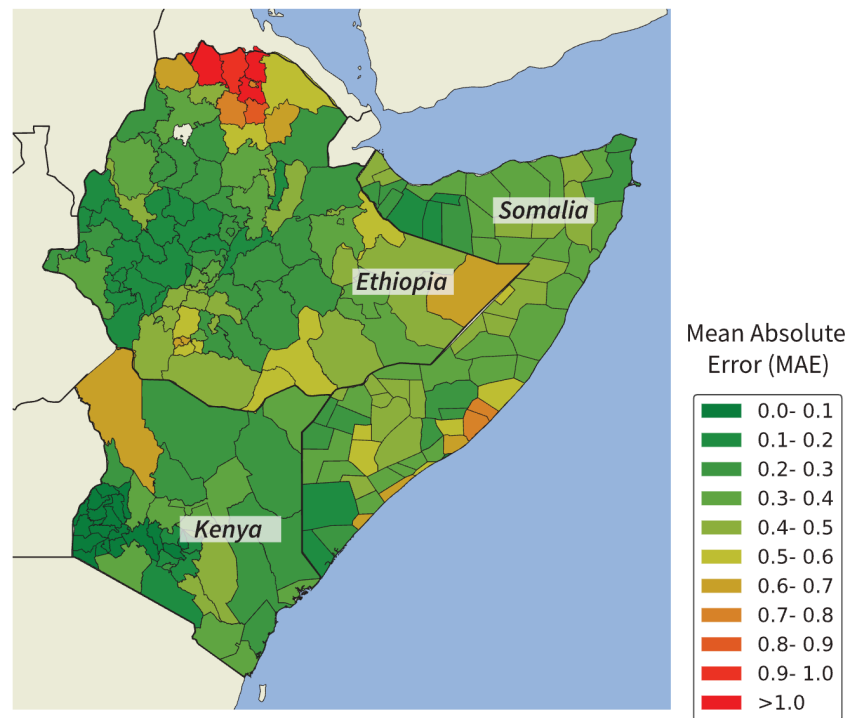


Figure 4. A spatial map of the mean absolute error (MAE) for the XGBoost machine-learning predictions of FEWS IPC food security, predicted 3 months ahead of the observation. Metrics are calculated from the test set (2019–2022) for all 213 administrative units.

related to the low variance in food security in such regions compared to agro-pastoral or pastoral regions. This low variance increases the relative capabilities of the persistence model (Figure 5, black line) for short leads and of the seasonality model (Figure 5, magenta line) for long leads. Note that the R^2 of the seasonality model is negative for the agro-pastoral and pastoral regions, which indicates that FEWS IPC seasonality does not generate any prediction skill. This outcome suggests that the FEWS IPC dynamics in the train data set (2009–2019) are notably different than those in the test data set (2019–2022) and emphasizes the need for more sophisticated predictions.

Subsequently, we compared the XGBoost forecasts to the FEWS NET food-security outlooks. These cutting-edge outlooks are produced by the FEWS NET early warning system and are widely used. The assessment suggests that the XGBoost model has similar performance to the FEWS NET outlooks in the agro-pastoral and pastoral regions. Compared to the FEWS NET outlooks, the skill of the XGBoost predictions seems to reduce less quickly for longer leads. However, for crop-farming regions, the FEWS NET outlooks perform considerably better, particularly for lead times longer than 3 months, where the R^2 of the XGBoost forecasts drops to below 0.5, but the FEWS NET outlooks remain well above 0.5.

3.2. Crisis-Onset Predictions

The above results indicate that general trends in food-security dynamics are captured by the XGBoost model. To use such a model as an early warning system, it is crucial to know to what extent the onset of a food crisis can be predicted. A food crisis onset has been defined as the transition from non-crisis (FEWS IPC <3) to crisis (FEWS IPC ≥ 3). Thus, in case a region stays in food crisis for multiple sequential time steps, only the first time step is counted as the crisis onset. In some cases, crises in certain administrative units were forecasted to start earlier (in the previous timestep) than they actually did. Although these forecasts accurately anticipated a crisis, they were deemed incorrect because the predicted onset was too early. We calculated the hit rate (number of correctly predicted crisis onsets/total number of crisis onsets) and the false-alarm rate (number of false alarms/number of no crisis onsets) in the test data set.

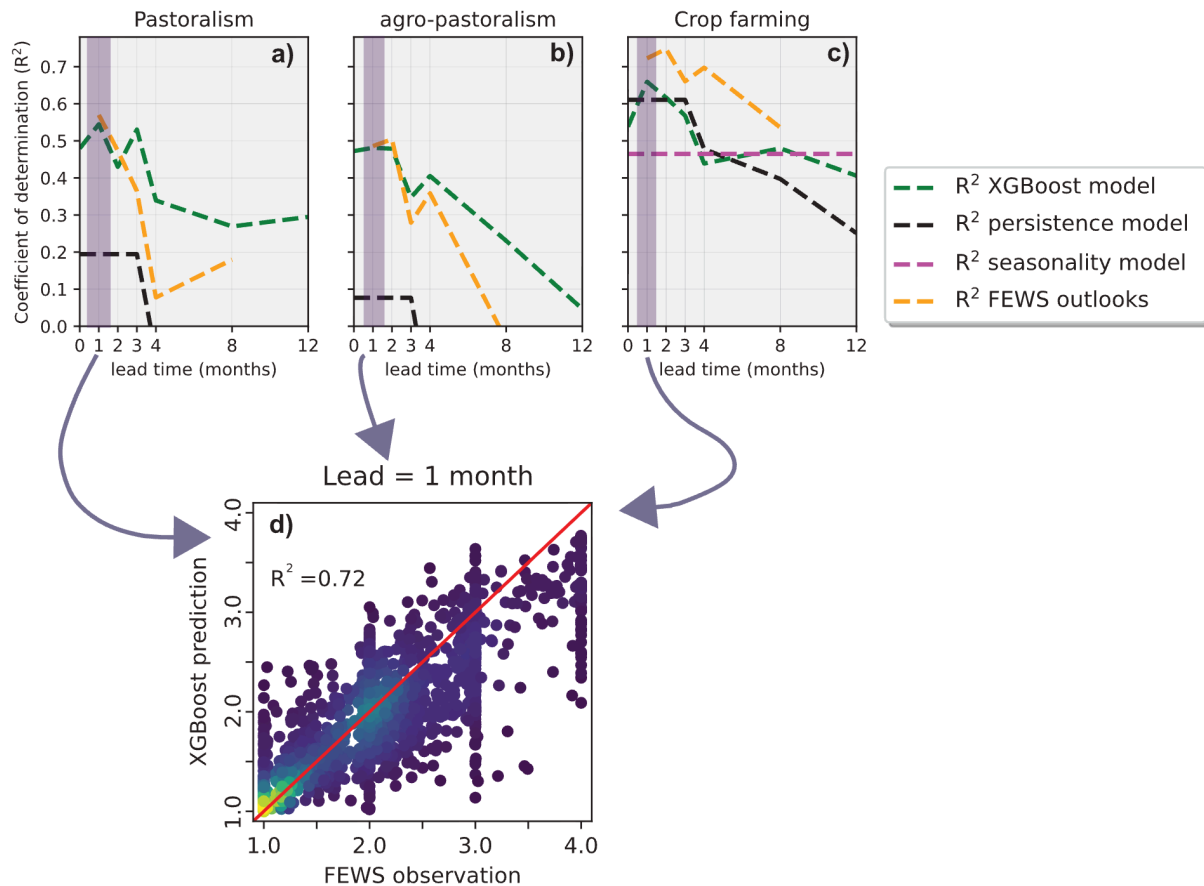


Figure 5. Top: The coefficient of determination (R^2) over lead time for the XGBoost predictions of food security in pastoral (a), agro-pastoral (b) and crop-farming (c) livelihoods are compared to the FEWS outlooks, the persistence model, and the seasonality model. R^2 values < 0 are not shown. Subsequently, the seasonality model in pastoralism and agro-pastoralism areas is not visible. Bottom (d): Individual XGBoost predictions and observations are visualized as a scatter plot for a lead time of 1 month. Individual dots represent a prediction or observation for one of the 213 administrative units. Colors represent the density of dots (Gaussian kernel-density estimate).

Results are shown in Figure 6. We predicted 20% of the food-crisis onsets for pastoral regions (out of 96 onsets in total) and 20%–50% for agro-pastoral regions (out of 22 onsets in total) using predictions up to a 3-month lead time (Figure 6, top). Predictions for these regions showed a low number of false alarms (around 5%). Over all lead times, food-crisis onsets in Somalia were consistently predicted more effectively, followed by those in Ethiopia and Kenya (see Figure 6, bottom left, for a lead time of 3 months). We predicted $>25\%$ of all food-crisis onsets in Somalia 1 month in advance with a very low number of false alarms. Predictions with a lead time of 4 months generally showed low performance. The significant drop in skill at 4 months is probably caused by the fact that the FEWS IPC observation of the previous timestep is no longer available to the model 4 months in advance. The model did not predict the start of a food crisis in the crop-farming regions (35 onsets in total). A potential explanation for this is that certain drivers for these regions could not be included. For example, we included proxies for crop yield (e.g., cropland NDVI) but did not have access to crop-yield data with acceptable accuracy.

The results presented in Figure 6 imply that the ability to detect food crises of our XGBoost model is similar to the FEWS NET outlooks for agro-pastoral and pastoral livelihood regions. We obtained low false alarm rates (generally $<10\%$), which is lower than the FEWS NET outlooks on all lead times. However, the FEWS NET outlooks clearly outperform our model in crop-farming regions (hit rate $>25\%$ – 65% for lead times up to 3 months).

This difference in predictive power potentially relates to variations in the input data and dynamic forecast models used in FEWS NET. Specifically, FEWS NET utilizes the G20 Group on Earth Observations Global Agriculture Monitoring (GEOGLAM) crop monitor to generate their outlooks (Funk, Shukla, et al., 2019) as well as soil moisture predictions from the NHyFAS system (Arsenault et al., 2020). Additionally, seasonal weather forecasts

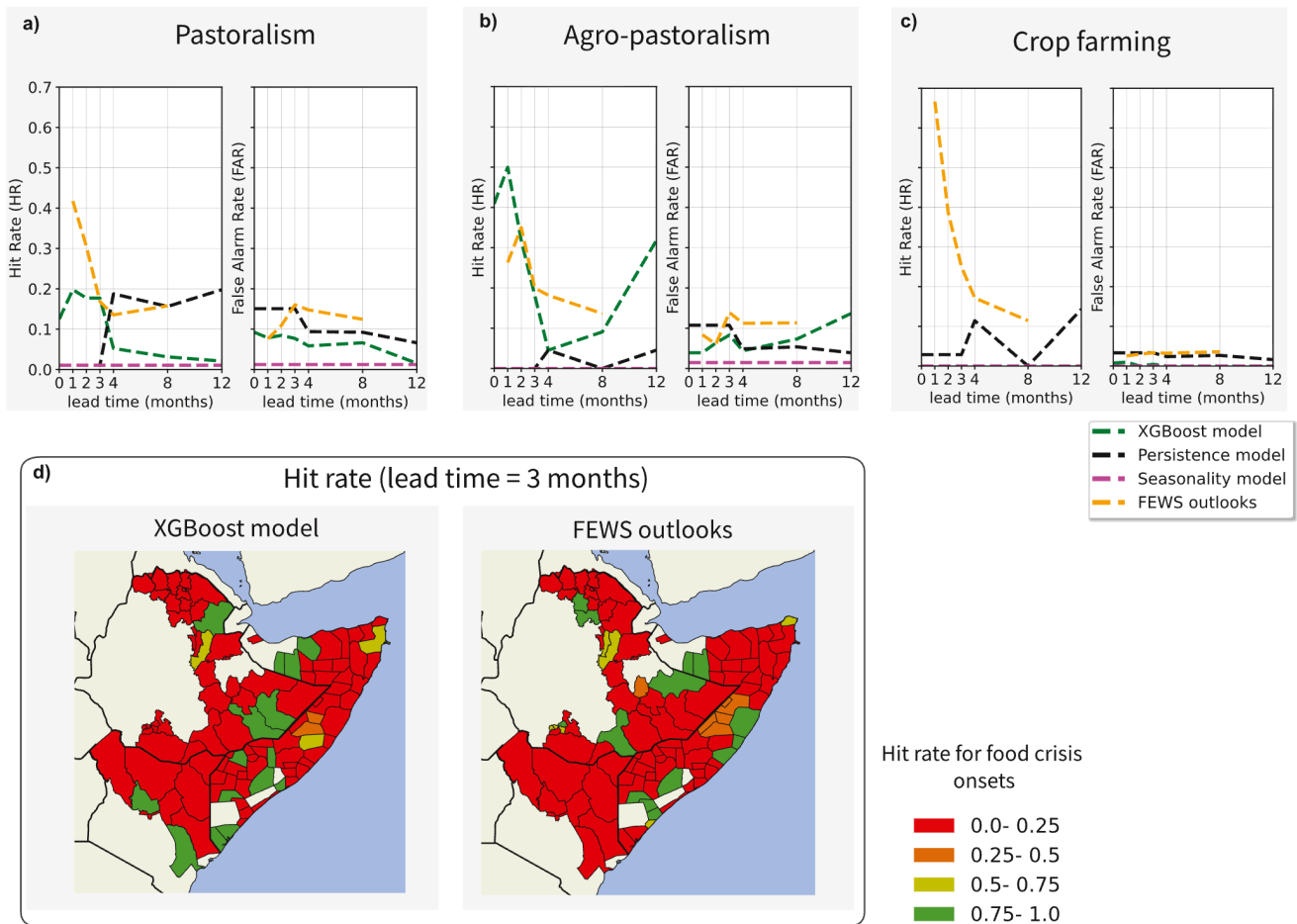


Figure 6. Top: The hit rate and false-alarm rate for the detection of crisis onsets (transition to FEWS IPC ≥ 3) are represented over multiple lead times for pastoralism (a), agro-pastoralism (b), and crop-farming (c) livelihoods. Bottom (d): A map of the hit rates for crisis onsets on lead 3 for the XGBoost model (left) and FEWS NET outlooks (right) is presented. Administrative units without any crisis observations in the test data set are masked. Metrics are based on the test period (2019–2022).

from NOAA and ICPAC are used in their outlooks. This data is not used in our model. Although the model evaluation is based on many crisis onsets, it reflects a relatively short period (2019–2022). Therefore, these results are indicative and do not have to reflect the true accuracy of the FEWS NET outlooks or the XGBoost model results.

3.3. Drivers of Food Insecurity

We assessed the impact of each feature on the predictions using the SHAP framework (Lundberg & Lee, 2017) for lead times of 3 and 8 months. To obtain insights into realistic potential food-insecurity drivers, we only show the best-performing administrative units (54 for lead 3 and 25 for lead 8). These are mostly located in (agro-)pastoral regions (Figure 7, bottom). We excluded regions with poor model performance because their key features may not be crucial local drivers of food security; instead, these drivers are simply not well-understood in these areas.

Food security is a consequence of multiple hazards (Boult et al., 2022), and it is well known that food security can have different potential drivers for different lead times (WMO, 2017). Therefore, we compared the importance of features in our models at different lead times. This resulted in a clear pattern (Figure 7). The previous FEWS NET food-security observation is highly important for predicting the next food-security status for short lead times (Figure 7a). However, this importance decreases at longer leads (Figure 7b), although it does not disappear entirely. Climate and weather variables—rainfall, evaporation (SPEI), and soil moisture (SSMI)—are relatively more important for predictability on short lead times (Figure 7a). These short lead-time drivers are in line with drivers found on the same lead time in Westerveld et al. (2021) for Ethiopia. In contrast, the predictability of long

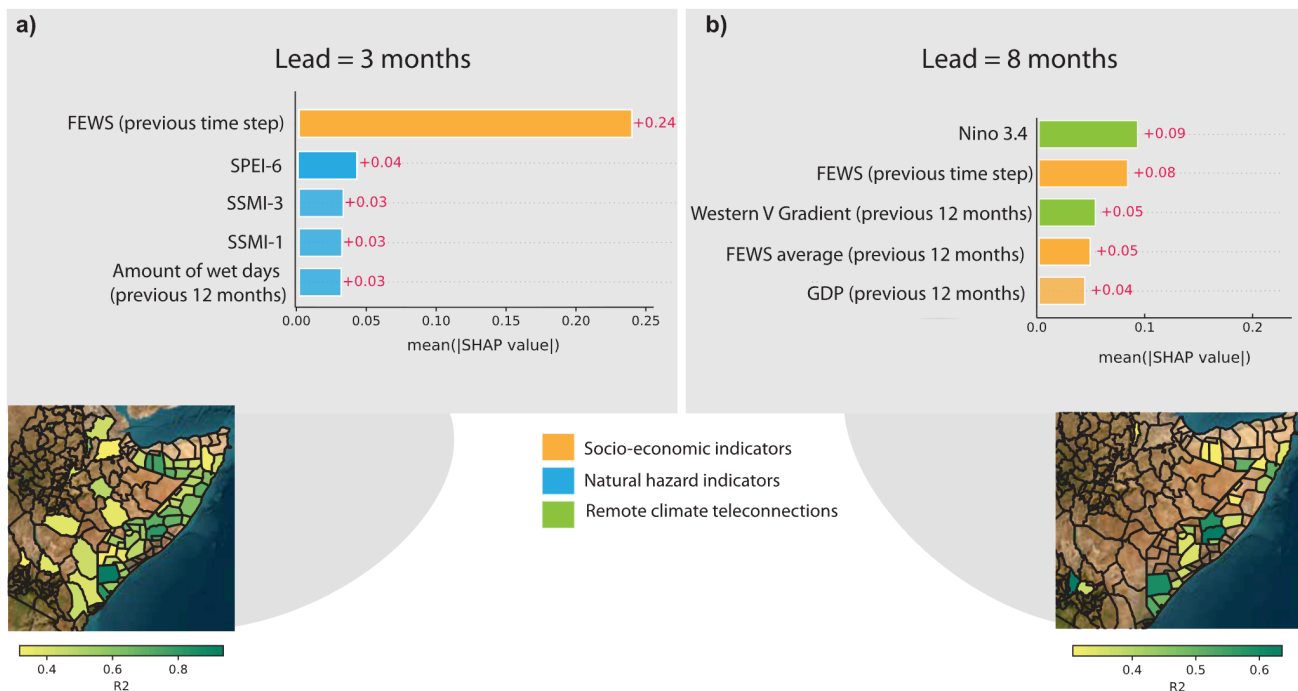


Figure 7. The top five most important features for a lead time of 3 months (a) and 8 months (b) for the best performing regions. Features are ranked based on the mean of the absolute SHAP values. Colors illustrate the different data categories (natural hazard, socio-economic, and remote climate teleconnections) in accordance with the colors in Figure 1. The maps (bottom) show the administrative units used for the SHAP calculation.

lead times (Figure 7b) mostly originates from remote climate teleconnections, such as Nino 3.4 and the WVG, as well as socio-economic indicators, such as the GDP. We found that the number of wet days per month is the most important rainfall indicator in the model, outnumbering SPI and the number of dry spells (Figure 7a). This suggests that food security has a stronger link with rainfall distribution over the month than with absolute rainfall amounts.

To understand the underlying model interactions, we further examined the relationship between model features and food insecurity. Figure 8 illustrates the impact of the 10 most important features in the model predictions on 4-month lead time for the 20% best performing administrative units. Nearly all the important natural hazard indicators have long accumulation periods: for example, SSMI-3 (Figure 8) and amount of wet days (Figure 7) use historical data over the last 3 and 12 months, respectively, to generate a forecast. This indicates that food security is mostly influenced by longer and more persistent drought conditions. Interestingly, the only variable that did not show these long accumulation periods was maize prices, which indicates that price spikes over a short period already had a strong effect on food security.

Apart from previous FEWS NET food-security states and maize prices, all selected features have a negative statistical relationship with food insecurity: lower feature values (blue dots, Figure 8) lead to increases in food insecurity (i.e., a positive impact on model output). This is to be expected, as drought (indicated by lower values of the SPI, SPEI, SSMI drought indices) can exacerbate or even trigger food insecurity (Funk, Shukla, et al., 2019).

The remote climate teleconnections also show an expected positive relationship with food insecurity (Figure 8): lower indices of Nino 3.4, WVG, IOD, or MEI are all drawing the East African climate toward a dryer state (Funk et al., 2023; Funk, Pedreros, et al., 2019). Negative IOD values imply relatively lower SSTs in the Western Indian Ocean, which result in less evaporating moisture transported into East Africa. Negative MEI and Nino 3.4 imply La Niña with overall cooler SSTs in the Eastern Pacific and warmer SSTs in the Western Pacific (Figure S1 in Supporting Information S1). The WVG reflects a warm blob in the Pacific Ocean around Indonesia and the Philippines (Figure S1 in Supporting Information S1), which have recently been found to be connected to the East African climate on long leads (Funk et al., 2023; Funk, Pedreros, et al., 2019). Our model results are consistent

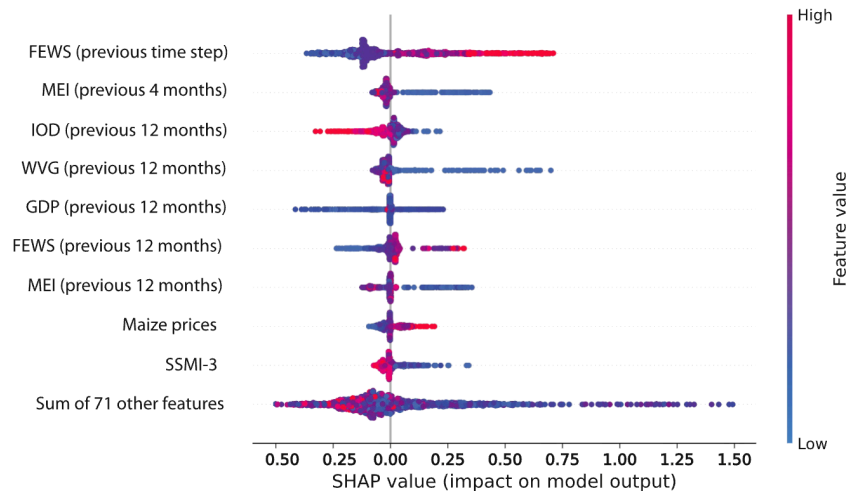


Figure 8. The top 10 most important features for model predictions with a lead time of 4 months, as indicated by the SHAP values for each feature's contribution. Each dot represents a prediction from the model, with the color indicating the value of the feature and the x-axis signifying whether the feature increased (positive SHAP values) or decreased (negative SHAP values) the model prediction of food insecurity. SHAP plots for all lead times are displayed in Figure S3 in Supporting Information S1.

with these findings and suggest that the WVG is also important for such impact-based forecasts, especially on long leads (Figure 7b). The SHAP values for the best performing regions found for the other lead times can be seen in Figure S3 in Supporting Information S1. These results demonstrate that the underlying model interactions are physically understandable and reflect intuition and the newest research insights.

4. Discussion

4.1. Adding Value to FEWS NET

This study indicates that machine-learning models can have similar performance to operational early warning systems such as FEWS NET. However, in crop-farming regions, the FEWS NET outlooks clearly outperformed the XGBoost model. This indicates that the machine-learning model could complement the existing FEWS NET outlooks in (agro-)pastoral regions, which are especially vulnerable to food crises. The FEWS NET outlooks are based on a scenario-development process and expert judgments, while our predictions are data-driven, enabling transparent generation of early warnings. Nonetheless, the FEWS NET current situation estimates are still a vital component of the machine-learning model and often the most important feature. This means that the expert assessments remain of high value in our machine-learning approach. Besides the more transparent generation of early warnings, another benefit is that potentially a higher initiation frequency can be achieved. Many of our utilized data sets are regularly updated, such as ACLED, which receives weekly updates, and the WPF VAM portal, which is updated biweekly. In practice, this could result in an increased frequency of outlook releases with relatively low additional financial resources and effort. Ultimately, this can result in more timely and improved early warnings. However, not all data sets are updated regularly and provide near real-time data. For example, the GLEAM data set is only updated once a year (GLEAM, 2024). In addition to complementing operational early warning systems, explainable AI techniques can reveal new potential food-security drivers for certain regions on lead times, which may then contribute to the scenario-development process for the FEWS NET food-security outlooks. A greater understanding of the drivers can also lead to more informed decisions on the ground and help tailor emergency-response planning.

These considerations demonstrate that such machine-learning predictions could add value as one component to operational systems, such as the FEWS NET early warning system. By no means can and should a machine-learning model entirely substitute early warning systems like FEWS NET. Relying solely on machine learning for disaster risk reduction is insufficient (Ghaffarian et al., 2023). On the ground expert judgment, local knowledge, and field experience are crucial to the co-production and functioning of early warning systems (ICPAC, 2021).

4.2. Machine-Learning Architecture: Key Considerations

The model results show that the prediction of acute food insecurity is complex, with many drivers that may contribute to changes in food-security status. Many different machine learning algorithms can be used to capture these drivers. We favored the use of a tree-based model over a neural-network model, as we did not expect strong multi-temporal or multi-spatial dependencies often found in other domains, such as image recognition (Fujiyoshi et al., 2019) or storm-surge modeling (Tiggeloven et al., 2021). Moreover, tree-based models often outperform neural networks on tabular data where features are individually meaningful (Lundberg et al., 2020).

We found that the XGBoost model was more capable of capturing these complex interactions than other tree-based models. We tested our results against random forest models, and the XGBoost model yielded a slightly higher performance, especially for crisis-onset predictions (see Figure S2 in Supporting Information S1). This is consistent with other studies. Westerveld et al. (2021) showed that the XGBoost model is superior to other machine-learning models for predicting transitions in food crises.

The feature engineering, which adds features based on existing ones, resulted in a total of 80 different features for the model. Some of these features are related (e.g., total precipitation over the last 4 and last 12 months). Feature selection—the reduction of features based on their (expected) performance—can help address multicollinearity (Chan et al., 2022), which is the correlation between predictor variables. While it can reduce multicollinearity, we opted not to perform feature selection in our study. This decision was influenced by XGBoost's robustness in handling multicollinearity. Additionally, we now retain all features in our 21 models, each tailored to different livelihood zones and lead times. Removing features risked losing vital insights specific to each model's context.

Before we pooled the data of the individual administrative units together (based on the three livelihood types), we tested many other spatial scales for model training. The levels involved were as follows: administrative units (21 individual models for each lead time), livelihood zones within all three countries, the three countries, and last, all data for the 213 administrative units pooled together with the livelihood zones as additional features. We found that increasing the level of spatial pooling significantly improved model performance. This can be explained by the limited period 2009–2019 used for training, which, within one administrative unit, simply does not contain sufficient data to learn the actual drivers of food insecurity. Pooling based on livelihood zones over the whole region performed most effectively, slightly better than pooling all data from every administrative unit together. Although the performance gain was small, making it difficult to draw definite conclusions, this could suggest that different livelihoods are influenced by distinct food-security drivers.

4.3. Limitations of the Study

4.3.1. The Data Sources

We created an extensive data set with food-security drivers from natural hazard indicators, socio-economic indicators, and remote-climate teleconnections. Although this is a holistic and extensive data set, certain important drivers and food insecurity precursors could not be included. We did not have access to crop-yield data on a monthly time scale for all the administrative units. Its exclusion may be a reason why the food-security crises could not be predicted effectively in crop-farming regions, whereas FEWS NET outlooks do predict these crises accurately (Figure 6). Krishnamurthy et al. (2022) found that incorporating additional remote sensing observations, such as those from the Soil Moisture Active Passive (SMAP) mission, could significantly enhance prediction capabilities. Balashankar et al. (2023) found that the inclusion of real-time news articles in a random forest model can lead to further improvements in the predictions of food crises. Future studies can expand on our research by including more of such data sources, which might lead to further model improvements. The performance of the model in detecting food crises in Kenya and Ethiopia is limited (Figure 6d). This, in part, can be explained by missing data. For example, maize prices—an important feature in our model—were not available for Kenya after 2020. Thus, although this feature can be used to train the model, it was missing during the model evaluation in the test set (2019–2022). Although the WFPs VAM portal provides price data for a high amount of markets, less markets were available for Kenya (14 markets) compared to Ethiopia (98) and Somalia (29). This issue emphasizes the importance of continuous data collection and archiving efforts. Future studies can include indicators reflecting market accessibility. While we only include the distance to the closest markets, the inclusion of market accessibility and functionality indicators (such as WFPs Market Functionality Index) could increase the model performance.

4.3.2. Model Testing

Although this study shows the potential of machine-learning systems for food-security early warning, the data could only be tested for a short period (2019–2022). This brings uncertainty to the verification of the predictions, especially for crisis onset events. Nonetheless, the test period of the study, 2019–2022, was highly suitable to test the predictions of food crisis onsets, as in this period many regions plunged into high levels of food insecurity. Therefore, we could evaluate our predictions on a total of 153 crisis onsets, making our skill estimates more robust.

The test period also reflects a turbulent period that, besides the 2020–2023 drought, encompassed the COVID-19 pandemic and the start of the Russian invasion of Ukraine. Both events have had significant economic effects worldwide, with economic fallout and price raises that in turn have resulted in record levels of malnutrition and food insecurity (WFP, 2022). Thus, food crises during the test period (2019–2022) were largely drought-driven, but embedded in conflict, the Russia-Ukraine conflict, and the COVID-19 pandemic that further complicated food security dynamics. Due to the unique characteristics of the test period, the model performance might change when tested on other periods. Future studies can test our models on different time periods.

4.3.3. The Role of Conflict in Predicting Food Crises

Conflicts are included in the model through the ACLED data set. However, they were not identified as an important variable in our model, as shown by its absence in the SHAP plots (Figures 8 and Figure S3 in Supporting Information S1). This is surprising, as research and practice show that conflict drives hunger (WFP, 2022). As such, we found that this missing link between conflict and hunger reduced the accuracy of our predictions. We failed to predict the rapid increase in food insecurity in the Tigray districts after the armed conflict erupted in November 2020 (Weldegiargis et al., 2023), which returned high error scores over this region for all lead times (MAE, Figure 4). Despite the ACLED data set showing clear signals in November 2020—three months prior to the crisis becoming apparent in the FEWS IPC data—our model did not anticipate the crisis at this lead time. This failure may be explained by the fact that the Tigray conflict was the largest recorded in the ACLED conflict data set over the 2009–2022 period, and therefore, the model could not train on conflicts of this magnitude. Moreover, some large conflicts in our data set (e.g., the Mogadishu bombings in October 2017) were followed by a decrease in food insecurity rather than an increase. As the source of error seems to be the model's missing link between conflict and food insecurity, integrating conflict forecasts into our model likely would not have enhanced the accuracy of our predictions for the Tigray food crisis. Conflict has consistently emerged as a complex factor in food-insecurity early warning, as highlighted in prior studies (Krishnamurthy et al., 2020, 2022). Interestingly, Backer and Billing (2024) found that conflict indicators are important features in their model forecasting the prevalence of child acute malnutrition. This could be explained by the inclusion of two additional conflict data sets besides ACLED: the Global Terrorism Database (LaFree & Dugan, 2007) and the UCDP Georeferenced Event Data set (Sundberg & Melander, 2013).

5. Conclusions and Recommendations

In this study, we developed an XGBoost machine-learning model to predict food insecurity on monthly timescales over the Horn of Africa. We trained the model on the 2009–2019 period using >20 different data sets and the FEWS IPC current situation as ground truth. Our model predicted 20% of crisis onsets in pastoral livelihood regions ($n = 96$) and 20%–50% of crisis onsets in agro-pastoral livelihood regions ($n = 22$) several months in advance with low overall numbers of false alarms. Furthermore, the model predicted general food-security patterns up to 3 months in advance ($R^2 > 0.6$). This underscores the potential of such machine-learning models to complement existing early warning systems, such as FEWS NET.

FEWS NET builds on decades of experience in food-security monitoring and early warning, and the FEWS NET food-security outlooks are widely adopted. To serve as an ultimate benchmark, we compared our predictions with these FEWS NET outlooks over the 2019–2022 period. Results suggest the performance of the XGBoost model is similar to the FEWS NET outlooks for agro-pastoral and pastoral regions. However, the FEWS NET outlooks clearly outperform our model for crop-farming regions. Moreover, machine-learning models need sufficient training data, which limited the data available during the test period (2019–2022). This decreased the robustness of model performance estimates. Thanks to continuing monitoring efforts from FEWS NET, more data will be available in the future to train and test such machine-learning models.

This study shows the potential of food-security predictions made with machine learning to complement existing early warning systems, such as FEWS NET, by allowing more frequent updates and revealing specific drivers in particular regions. Future research could further explore how such machine-learning models can be improved. We expect that the inclusion of dynamical forecasts as features in the machine-learning model will lead to a significant improvement. Soil moisture and yield forecasts (Shukla et al., 2020) can lead to better predictions in crop farming regions. Moreover, decision-makers can use results from this study as indications for potentially important drivers of food-security crises at different lead times, which may lead to more informed and timely interventions. The organizations operating and developing food security early warning systems can use our results to envision and shape hybrid solutions where a part is automated and based on machine learning, but also a part remains consensus-based.

Data Availability Statement

The input data used to run the food-security machine learning model in the study are available at Zenodo (Busker, 2023b) with the Creative Commons Attribution 4.0 International license. Version 1.0.1 of the machine learning model used to generate the food security predictions is preserved at Zenodo (Busker, 2023a), available with the Creative Commons Attribution 4.0 International license.

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References

- Arsenault, K. R., Shukla, S., Hazra, A., Getirana, A., McNally, A., Kumar, S. V., et al. (2020). The NASA hydrological forecast system for food and water security Applications. *Bulletin of the American Meteorological Society*, 101(7), E1007–E1025. <https://doi.org/10.1175/BAMS-18-0264.1>
- Backer, D., & Billing, T. (2021). Validating famine early warning systems network projections of food security in Africa, 2009–2020. *Global Food Security*, 29, 100510. <https://doi.org/10.1016/j.gfs.2021.100510>
- Backer, D., & Billing, T. (2024). Forecasting the prevalence of child acute malnutrition using environmental and conflict conditions as leading indicators. *World Development*, 176, 106484. <https://doi.org/10.1016/j.worlddev.2023.106484>
- Balashankar, A., Subramanian, L., & Fraiberger, S. P. (2023). Predicting food crises using news streams. *Science Advances*, 9(9), eabm3444. <https://doi.org/10.1126/sciadv.abm3449>
- Baxter, A. J., Verschuren, D., Peterse, F., Miralles, D. G., Martin-Jones, C. M., Maitiuerdi, A., et al. (2023). Reversed holocene temperature moisture relationship in the Horn of Africa. *Nature*, 620(7973), 336–343. <https://doi.org/10.1038/s41586-023-06272-5>
- Blauhut, V., Stahl, K., Stagge, J. H., Tallaksen, L. M., De Stefano, L., & Vogt, J. (2016). Estimating drought risk across Europe from reported drought impacts, drought indices, and vulnerability factors. *Hydrology and Earth System Sciences*, 20(7), 2779–2800. <https://doi.org/10.5194/hess-20-2779-2016>
- Boult, V. L., Asfaw, D. T., Young, M., Maidment, R., Mwangi, E., Ambani, M., et al. (2020). Evaluation and validation of TAMSAT-ALERT soil moisture and WRSI for use in drought anticipatory action. *Meteorological Applications*, 27(5), 27. <https://doi.org/10.1002/met.1959>
- Boult, V. L., Black, E., Saado Abdullahi, H., Bailey, M., Harris, C., Kilavi, M., et al. (2022). Towards drought impact-based forecasting in a multiple hazard context. *Climate Risk Management*, 35, 100402. <https://doi.org/10.1016/j.crm.2022.100402>
- Busker, T. (2023a). Model scripts used for the paper “predicting food-security crises in the Horn of Africa using machine learning” (version v1.0.1). [Software]. Zenodo. <https://doi.org/10.5281/zenodo.10853621>
- Busker, T. (2023b). Replication data for: “Predicting food-security crises in the Horn of Africa using machine learning” (version v1.0.1) [Dataset]. Zenodo. <https://doi.org/10.5281/zenodo.10853668>
- Cerqueira, V., Torgo, L., & Mozetič, I. (2020). Evaluating time series forecasting models: An empirical study on performance estimation methods. *Machine Learning*, 109(11), 1997–2028. <https://doi.org/10.1007/s10994-020-05910-7>
- Chan, J. Y.-L., Leow, S. M. H., Bea, K. T., Cheng, W. K., Phoong, S. W., Hong, Z.-W., & Chen, Y.-L. (2022). Mitigating the multicollinearity problem and its machine learning approach: A review. *Mathematics*, 10(8), 1283. <https://doi.org/10.3390/math10081283>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 785–794). ACM. <https://doi.org/10.1145/2939672.2939785>
- Dempsey, B., & Hillier, D. (2012). A Dangerous Delay: The cost of late response to early warnings in the 2011 drought in the Horn of Africa. Retrieved from <https://policy-practice.oxfam.org/resources/a-dangerous-delay-the-cost-of-late-response-to-early-warnings-in-the-2011-droug-203389/>
- Everingham, Y., Sexton, J., Skocaj, D., & Inman-Bamber, G. (2016). Accurate prediction of sugarcane yield using a random forest algorithm. *Agronomy for Sustainable Development*, 36(2), 27. <https://doi.org/10.1007/s13593-016-0364-z>
- FAO. (2009). Declaration of the world summit on food security (WSFS 2009/2). Retrieved from <https://reliefweb.int/report/world/declaration-world-summit-food-security-wsfs-20092>
- FAO. (2022). FAO locust Hub | swarms. [Dataset]. FAO. Retrieved from <https://locust-hub-hqfao.hub.arcgis.com/datasets/hqfao:swarms-explore>
- FAO. (2023a). GIEWS - global information and early warning system - country briefs. Retrieved from <https://www.fao.org/giews/countrybriefs/index.jsp>
- FAO. (2023b). GIEWS - global information and early warning system on food and agriculture. Retrieved from <https://www.fao.org/giews/en/>
- FEWS NET. (2018). Scenario development for food security early warning. Retrieved from https://fewsn.net/sites/default/files/documents/reports/Guidance_Document_Scenario_Development_2018.pdf
- FEWS NET. (2023a). FEWS NET data explorer. [Dataset]. FEWS NET. Retrieved from <https://fdw.fews.net/en/>
- FEWS NET. (2023b). Monitoring & forecasting acute food insecurity. Retrieved from <https://fewsn.net/>
- FEWS NET. (2024). What is the IPC? | FEWS NET. Retrieved from <https://fewsn.net/about/integrated-phase-classification>

- Foini, P., Tizzoni, M., Martini, G., Paolotti, D., & Omodei, E. (2023). On the forecastability of food insecurity. *Scientific Reports*, 13(1), 2790. <https://doi.org/10.1038/s41598-023-29700-y>
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232. <https://doi.org/10.1214/aos/1013203451>
- Fujiyoshi, H., Hirakawa, T., & Yamashita, T. (2019). Deep learning-based image recognition for autonomous driving. *IATSS Research*, 43(4), 244–252. <https://doi.org/10.1016/j.iatssr.2019.11.008>
- Funk, C., Harrison, L., Segele, Z., Rosenstock, T., Steward, P., Anderson, C. L., et al. (2023). Tailored forecasts can predict extreme climate events informing proactive interventions in East Africa. *Earth's Future*, 11(7), e2023EF003524. <https://doi.org/10.1029/2023EF003524>
- Funk, C., Pedreros, D., Nicholson, S., Hoell, A., Korecha, D., Galu, G., et al. (2019). Examining the potential contributions of extreme “Western V” sea surface temperatures to the 2017 March–June East African drought. *Bulletin of the American Meteorological Society*, 100(1), S55–S66. <https://doi.org/10.1175/BAMS-D-18-0108.1>
- Funk, C., Peterson, P., Landsfeld, M., Pedreros, D., Verdin, J., Shukla, S., et al. (2015). The climate hazards infrared precipitation with stations—A new environmental record for monitoring extremes. *Scientific Data*, 2(1), 1–21. <https://doi.org/10.1038/SDATA.2015.66>
- Funk, C. & Shukla, S. (Eds.) (2020). *Drought early warning and forecasting - theory and practice* (pp. 215–222). Elsevier. <https://doi.org/10.1016/B978-0-12-814011-6.00020-8>
- Funk, C., Shukla, S., Thiaw, W. M., Rowland, J., Hoell, A., McNally, A., et al. (2019). Recognizing the famine early warning systems network over 30 years of drought early warning science advances and partnerships promoting global food security. *Bulletin of the American Meteorological Society*, 100(6), 1011–1027. <https://doi.org/10.1175/BAMS-D-17-0233.1>
- Ghaffarian, S., Taghikhah, F. R., & Maier, H. R. (2023). Explainable artificial intelligence in disaster risk management: Achievements and prospective futures. *International Journal of Disaster Risk Reduction*, 98, 104123. <https://doi.org/10.1016/j.ijdrr.2023.104123>
- GLEAM. (2024). GLEAM | global land evaporation Amsterdam model. Retrieved from <https://www.gleam.eu/>
- Guimarães Nobre, G., Davenport, F., Bischiniotis, K., Veldkamp, T., Jongman, B., Funk, C. C., et al. (2019). Financing agricultural drought risk through ex-ante cash transfers. *Science of the Total Environment*, 653, 523–535. <https://doi.org/10.1016/j.scitotenv.2018.10.406>
- Ha, J., Kose, M. A., & Ohnsorge, F. (2021). One-stop source: A global database of inflation. *Policy Research*. Working Paper 9737.
- Hao, Z., AghaKouchak, A., Nakhjiri, N., & Farahmand, A. (2014). Global integrated drought monitoring and prediction system. *Scientific Data*, 1(1), 140001. <https://doi.org/10.1038/sdata.2014.1>
- ICPAC. (2021). Co-Production in climate services: What, why, and how? Retrieved from <https://www.icpac.net/news/co-production-in-climate-services-what-why-and-how/>
- IMF. (2023). World economic outlook (April 2023). [Dataset]. IMF. Retrieved from <https://www.imf.org/external/datamapper/datasets/WEO/>
- IPC. (2021). IPC technical manual version 3.1. Retrieved from <http://www.ipcinfo.org/ipc-manual-interactive/en/>
- IPC. (2023). IPC overview and classification system | IPC - integrated food security phase classification. Retrieved from <https://www.ipcinfo.org/ipcinfo-website/ipc-overview-and-classification-system/en/>
- Kimutai, J., Barnes, C., Zachariah, M., Philip, S., Kew, S., Pinto, I., et al. (2023). Human-induced climate change increased drought severity in the Horn of Africa. Imperial College London. <https://doi.org/10.25561/103482>
- Krishnamurthy, P. K., Choularton, R. J., & Kareiva, P. (2020). Dealing with uncertainty in famine predictions: How complex events affect food security early warning skill in the Greater Horn of Africa. *Global Food Security*, 26, 100374. <https://doi.org/10.1016/j.gfs.2020.100374>
- Krishnamurthy, P. K., Fisher, J. B., Choularton, R. J., & Kareiva, P. M. (2022). Anticipating drought-related food security changes. *Nature Sustainability*, 5(11), 956–964. <https://doi.org/10.1038/s41893-022-00962-0>
- LaFree, G., & Dugan, L. (2007). Introducing the global terrorism database. *Terrorism and Political Violence*, 19(2), 181–204. <https://doi.org/10.1080/09546550701246817>
- Lee, D., Davenport, F., Shukla, S., Husak, G., Funk, C., Harrison, L., et al. (2022). Maize yield forecasts for Sub-Saharan Africa using Earth Observation data and machine learning. *Global Food Security*, 33, 100643. <https://doi.org/10.1016/j.gfs.2022.100643>
- Lentz, E., & Maxwell, D. (2022). How do information problems constrain anticipating, mitigating, and responding to crises? *International Journal of Disaster Risk Reduction*, 81, 103242. <https://doi.org/10.1016/j.ijdrr.2022.103242>
- Lentz, E., Michelson, H., Baylis, K., & Zhou, Y. (2019). A data-driven approach improves food insecurity crisis prediction. *World Development*, 122, 399–409. <https://doi.org/10.1016/j.worlddev.2019.06.008>
- Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., et al. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1), 56–67. <https://doi.org/10.1038/s42256-019-0138-9>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions (version 2). <https://doi.org/10.48550/ARXIV.1706.03522>
- Martens, B., Miralles, D. G., Lievens, H., van der Schalie, R., de Jeu, R. A. M., Fernández-Prieto, D., et al. (2017). GLEAM v3: Satellite-based land evaporation and root-zone soil moisture. *Geoscientific Model Development*, 10(5), 1903–1925. <https://doi.org/10.5194/gmd-10-1903-2017>
- Martini, G., Bracci, A., Riches, L., Jaiswal, S., Corea, M., Rivers, J., et al. (2022). Machine learning can guide food security efforts when primary data are not available. *Nature Food*, 3(9), 716–728. <https://doi.org/10.1038/s43016-022-00587-8>
- McGovern, A., Lagerquist, R., Gagne, D. J., Jergensen, G. E., Elmore, K. L., Homeyer, C. R., & Smith, T. (2019). Making the black box more transparent: Understanding the physical implications of machine learning. *Bulletin of the American Meteorological Society*, 100(11), 2175–2199. <https://doi.org/10.1175/BAMS-D-18-0195.1>
- McKee, T. B., Doesken, N. J., & Kleist, J. (1993). The relationship of drought frequency and duration to time scales. *Proceedings of the 8th Conference on Applied Climatology*, 17(22).
- NBS. (2023). Somali national Bureau of Statistics. [Dataset]. NBS. Retrieved from <https://www.nbs.gov.so/home>
- NOAA. (2023a). Climate indices: Monthly atmospheric and ocean time series. [Dataset]. NOAA. Retrieved from <https://psl.noaa.gov/data/climateindices/list/>
- NOAA. (2023b). Download climate timeseries - Dipole mode index (DMI). [Dataset]. NOAA. Retrieved from https://psl.noaa.gov/gcos_wgsp/Timeseries/DMI/
- NOAA. (2021). Star - global vegetation health products: Downloading vegetation health products data. [Dataset]. NOAA. Retrieved from <https://www.star.nesdis.noaa.gov/smcd/emb/vci/VH/vhftp.php>
- Odongo, R. A., De Moel, H., & Van Loon, A. F. (2023). Propagation from meteorological to hydrological drought in the Horn of Africa using both standardized and threshold-based indices. *Natural Hazards and Earth System Sciences*, 23(6), 2365–2386. <https://doi.org/10.5194/nhess-23-2365-2023>
- Pérez-Hoyos, A. (2018). Global crop and rangeland masks. European Commission. [Dataset]. Joint Research Centre (JRC). Retrieved from <http://data.europa.eu/89h/jrc-10112-10005>

- Raleigh, C., Linke, R., Hegre, H., & Karlsen, J. (2010). Introducing ACLED: An armed conflict location and event dataset. *Journal of Peace Research*, 47(5), 651–660. <https://doi.org/10.1177/0022343310378914>
- Schoppa, L., Disse, M., & Bachmair, S. (2020). Evaluating the performance of random forest for large-scale flood discharge simulation. *Journal of Hydrology*, 590, 125531. <https://doi.org/10.1016/j.jhydrol.2020.125531>
- Shapley, L. S. (1953). A value for n-person games. In H. W. Kuhn & A. W. Tucker (Eds.), *Contributions to the theory of games (AM-28)* (Vol. pp. 307–318). Princeton University Press. <https://doi.org/10.1515/9781400881970-018>
- Shukla, S., Arsenault, R., Hazra, A., Peters-Lidard, C., Koster, R., Davenport, F., et al. (2020). Improving early warning of drought-driven food insecurity in southern Africa using operational hydrological monitoring and forecasting products. *Natural Hazards and Earth System Science*, 20(4), 1187–1201. <https://doi.org/10.5194/NHESS-20-1187-2020>
- Snijders, T. A. B. (1988). On cross-validation for predictor evaluation in time series. In T. K. Dijkstra (Ed.), *On model uncertainty and its statistical implications* (Vol. 307, pp. 56–69). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-61564-1_4
- Stagge, J. H., Kohn, I., Tallaksen, L. M., & Stahl, K. (2015). Modeling drought impact occurrence based on meteorological drought indices in Europe. *Journal of Hydrology*, 530, 37–50. <https://doi.org/10.1016/j.jhydrol.2015.09.039>
- Sundberg, R., & Melander, E. (2013). Introducing the UCDP georeferenced event dataset. *Journal of Peace Research*, 50(4), 523–532. <https://doi.org/10.1177/0022343313484347>
- Svoboda, M. D., & Fuchs, B. A. (2017). Handbook of drought indicators and indices. In *Drought and water crises: Integrating science, management, and policy* (2nd ed.). <https://doi.org/10.1201/b22009>
- Tiggeloven, T., Couasnon, A., van Straaten, C., Muis, S., & Ward, P. J. (2021). Exploring deep learning capabilities for surge predictions in coastal areas. *Scientific Reports*, 11(1), 1–15. <https://doi.org/10.1038/s41598-021-96674-0>
- UNHCR. (2023). Horn of Africa food crisis explained. Retrieved from <https://www.unrefugees.org/news/horn-of-africa-food-crisis-explained/#%20Why%20is%20there%20food%20insecurity%20in%20the%20Horn%20of%20Africa?>
- Vicente-Serrano, S. M., Beguería, S., & López-Moreno, J. I. (2010). A multiscale drought index sensitive to global warming: The standardized precipitation evapotranspiration index. *Journal of Climate*, 23(7), 1696–1718. <https://doi.org/10.1175/2009JCLI2909.1>
- Wang, D., Andree, B. P. J., Chamorro, A. F., & Girouard Spencer, P. (2020). *Stochastic modeling of food insecurity*. World Bank. <https://doi.org/10.1596/1813-9450-9413>
- Weldegiargis, A. W., Abebe, H. T., Abraha, H. E., Abrha, M. M., Tesfay, T. B., Belay, R. E., et al. (2023). Armed conflict and household food insecurity: Evidence from war-torn Tigray, Ethiopia. *Conflict and Health*, 17(1), 22. <https://doi.org/10.1186/s13031-023-00520-1>
- Westerveld, J. J. L., van den Homberg, M. J. C., Nobre, G. G., van den Berg, D. L. J., Teklesadik, A. D., & Stuit, S. M. (2021). Forecasting transitions in the state of food security with machine learning using transferable features. *Science of the Total Environment*, 786, 147366. <https://doi.org/10.1016/j.scitotenv.2021.147366>
- WFP. (2014). Technical guidance note: Calculation and use of the alert for price spikes (ALPS) indicator. Retrieved from https://documents.wfp.org/stellent/groups/public/documents/manual_guide_proced/wfp264186.pdf
- WFP. (2022). *War in Ukraine drives global food crisis* | World Food Programme. Retrieved from <https://www.wfp.org/publications/war-ukraine-drives-global-food-crisis>
- WFP. (2023a). Scaling up anticipatory actions for food security: Anticipatory action year in focus 2022. Retrieved from <https://www.wfp.org/publications/scaling-anticipatory-actions-food-security-anticipatory-action-year-focus-2022>
- WFP. (2023b). VAM food security analysis -economic explorer. [Dataset]. WFP. Retrieved from <https://dataviz.vam.wfp.org/economic-explorer/prices>
- WFP and FAO. (2022). Hunger Hotspots FAO-WFP early warnings on acute food insecurity: October 2022 to January 2023 Outlook. Retrieved from <https://docs.wfp.org/api/documents/WFP-0000142656/download/>
- WMO. (2017). How climate forecasts strengthen food security. Retrieved from <https://public.wmo.int/en/resources/bulletin/how-climate-forecasts-strengthen-food-security>
- WMO. (2022). Meteorological and humanitarian agencies sound alert on East Africa. Retrieved from <https://public.wmo.int/en/media/news/meteorological-and-humanitarian-agencies-sound-alert-east-africa>
- WorldPop. (2024). WorldPop: Population density. [Dataset]. WorldPop. Retrieved from <https://hub.worldpop.org/geodata/listing?id=77>
- Zhang, B., Abu Salem, F. K., Hayes, M. J., Smith, K. H., Tadesse, T., & Wardlow, B. D. (2023). Explainable machine learning for the prediction and assessment of complex drought impacts. *Science of the Total Environment*, 898, 165509. <https://doi.org/10.1016/j.scitotenv.2023.165509>
- Zheng, A., & Casari, A. (2018). *Feature engineering for machine learning: Principles and techniques for data scientists*. O'Reilly Media, Inc.